Chapter 6: Additional Forces

The following subjects will be discussed

- Friction.
- Drag.
- Circular motion.

Friction

The surface roughness presents some kind of resistance to motion called friction.

Friction magnitude is proportional to the normal force (not gravitational force).

Friction direction is opposite to the direction of motion.

Static friction: prevents stationary objects from moving or sliding over surfaces.

$$f_s = \mu_s \cdot F_N$$

Kinetic friction: impedes moving/sliding objects trying to stop them.

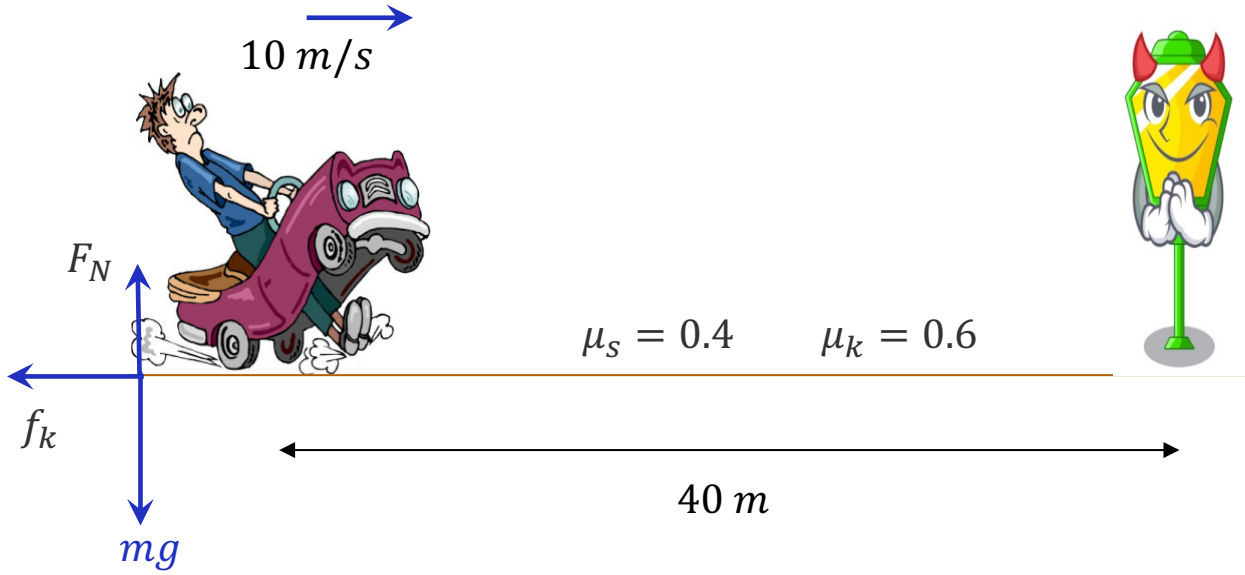
$$f_k = \mu_k \cdot F_N$$

The coefficients μ_s and μ_k are dimensionless and depend on the surface.

Rolling is not considered here!

Examples

1) Will we see a crash??



$$\vec{F}_{net} = m\,\vec{a}$$

$$-f_k = m\,a$$

$$-f_k = -\mu_k F_N = -\mu_k mg = m\,a$$

$$a = -\mu_k g$$

$$a = -5.88\ m/s^2$$

$$v^2 = v^2{}_o + 2a(x - x_o)$$

$$0 = 10^2 + 2(-5.88)\,d$$

$$d = 8.5\ m$$

On an icy day

$$\mu_k = 0.1$$

$$d = 51\ m$$

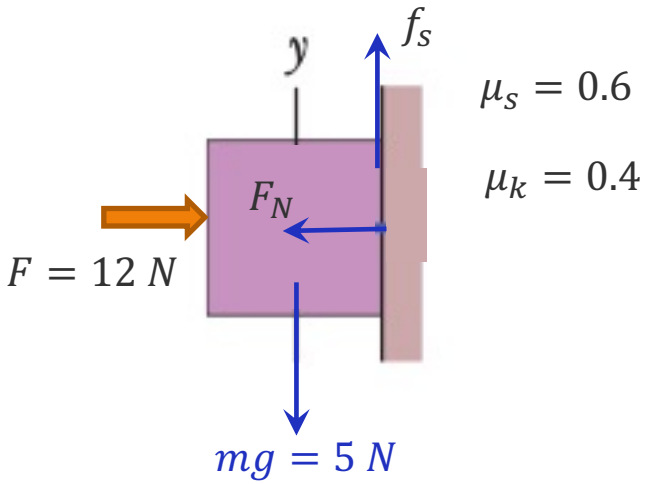
Examples

2) Will the block slide??

$$f_{s_max} = \mu_s F_N$$

$$= (0.6)(12)$$

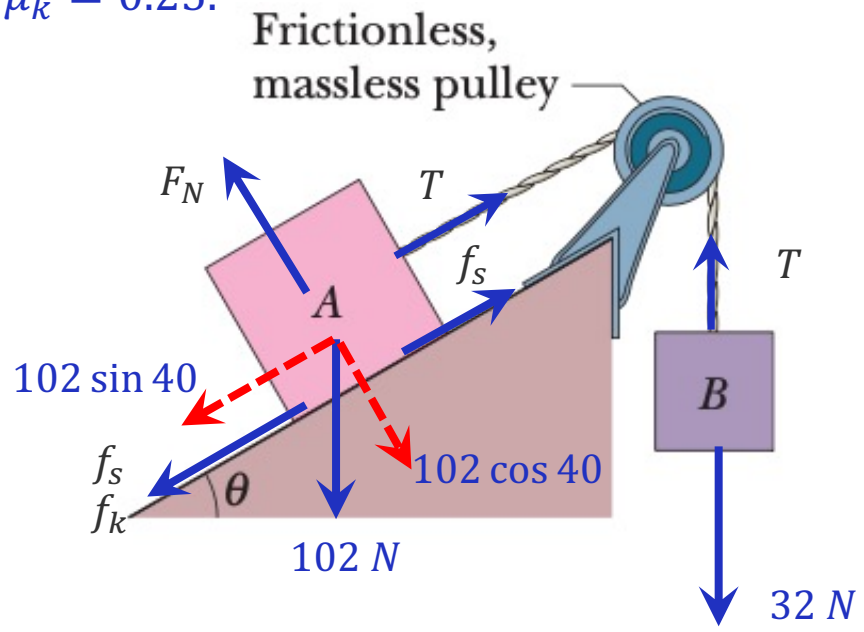
$$= 7.2\ N > 5\ N$$



No!

Examples

3) Find the acceleration. $\theta = 40°$, $\mu_s = 0.56$, $\mu_k = 0.25$.



If the system is initially at rest. $T = 32\ N$

Will it move up? No

$mg \sin 40 + f_s \quad ? \quad T$

$102 \sin 40 + 0.56(102 \cos 40) \quad > \quad 32$

Will it move down? No

$mg \sin 40 \quad ? \quad T + f_s$

$102 \sin 40 \quad < \quad 32 + 0.56(102 \cos 40)$

Will not move at all.

If the system is initially moving up.

$T - 102 \sin 40 - f_k = m_A a$

$32 - T = m_B a$

Add the two equations and solve for a

$a = -3.9\ \mathrm{m/s^2}$

Drag

Objects penetrating through fluids experience drag. (Or fluid moving past object)

Drag force magnitude depends on 1) object properties, 2) fluid and 3) speed.

Turbulent Flow

$$D = \frac{1}{2} C \rho A v^2$$

- C: Fluid drag coefficient (0.4-1.0)
- ρ: Fluid density
- A: Object effective cross-sectional area
- v: Object speed

Drag force direction is opposite to motion.

As an object free-falls, it is initially accelerated downward with g. As the velocity increases, D increases. A condition of $F_g - D = 0$ can be reached.

The speed becomes constant (terminal speed)

$$v_t = \sqrt{\frac{2F_g}{C \rho A}}$$

Examples

1) Study the effect of air drag on rain drops.



Air

R=1.5 mm

ρ_{water}=1000 kg/m^3

C=0.6

ρ_{air}=1.2 kg/m^3



1200 m

$$F_g = mg = V \cdot \rho_{water} \cdot g = \frac{4}{3}\pi R^3 \cdot \rho_{water} \cdot g$$

$$A = \pi R^2$$

$$v_t = \sqrt{\frac{2F_g}{C\,\rho\,A}} = 7.4 \text{ m/s} \quad (27 \text{ km/hr})$$

Without drag $\quad v = \sqrt{2gh} = 550$ km/hr

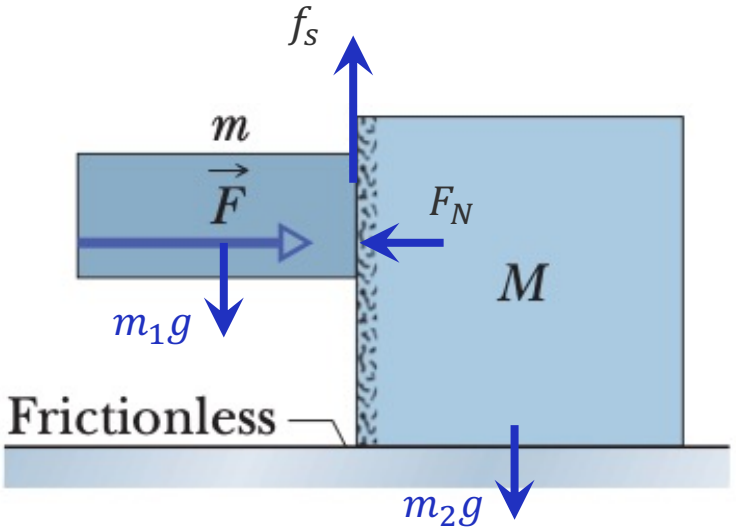
Examples

2) What is the minimum force F such that mass m_1 will not slide down?

The condition for minimum force is

$$m_1 g - f_s = 0$$

$$m_1 g - \mu_s F_N = 0 \quad (1)$$



Let's find F_N.

$$F = (m_1 + m_2)a \quad (2)$$

$$F - F_N = m_1 a \quad (3)$$

Divide 2/3 and solve for F_N.

$$F_N = \frac{m_2}{m_1 + m_2} F$$

Sub. In 1

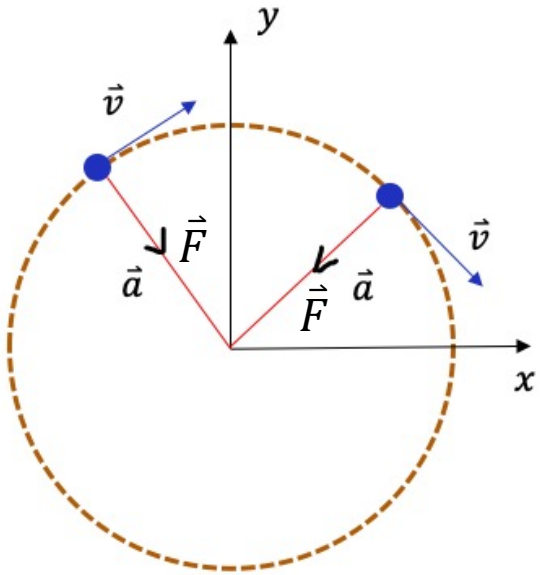
$$m_1 g - \mu_s \frac{m_2}{m_1 + m_2} F = 0$$

$$F = \frac{m_1 (m_1 + m_2) g}{\mu_s m_2} = 488\ N$$

Uniform Circular Motion

There is a central force associated with circular motion.

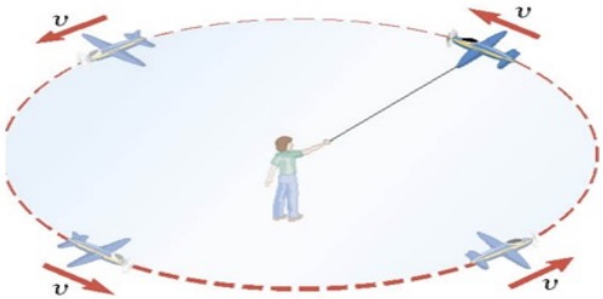
This force is connected to gravity, tension, etc.



$$\vec{a} = \frac{v^2}{r}\,\vec{a}_n$$

$$\vec{F} = m\vec{a}$$

$$|F| = \frac{mv^2}{r}$$



$\vec{v}$ is tangential to path.

$\vec{F}$ and $\vec{a}$ change direction continuously.

If $\vec{F} = 0$, the object moves in a straight line.

$\vec{a}$ is linear with radius.

The equations also apply for partial circular paths.

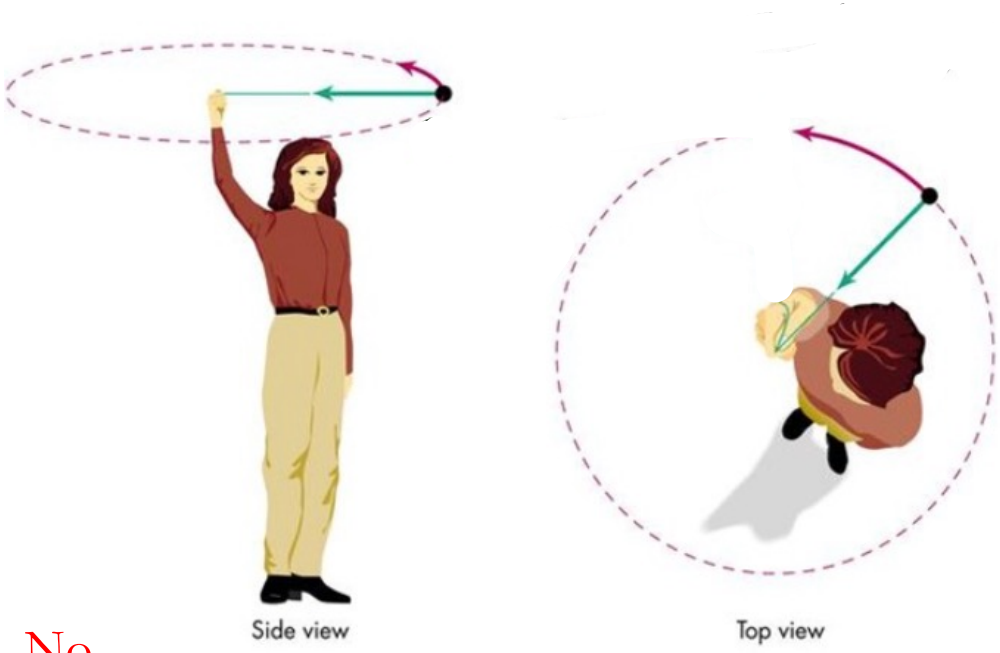
Examples

1) Spinning ball. Find the maximum velocity before the cord snaps.

$$T_{max} = F_{cent} = \frac{m}{R} v_{max}^2$$

$$v_{max} = \sqrt{\frac{T_{max}\, R}{m}}$$

$$v_{max} = \sqrt{\frac{(50)(1.5)}{0.5}} = 12.2\ m/s$$



If R is doubled, will the cord snap? No

$$\frac{T_1}{T_2} = \frac{\frac{mv_1^2}{R_1}}{\frac{mv_2^2}{R_2}} \Rightarrow T_2 = \frac{T_1}{2}$$

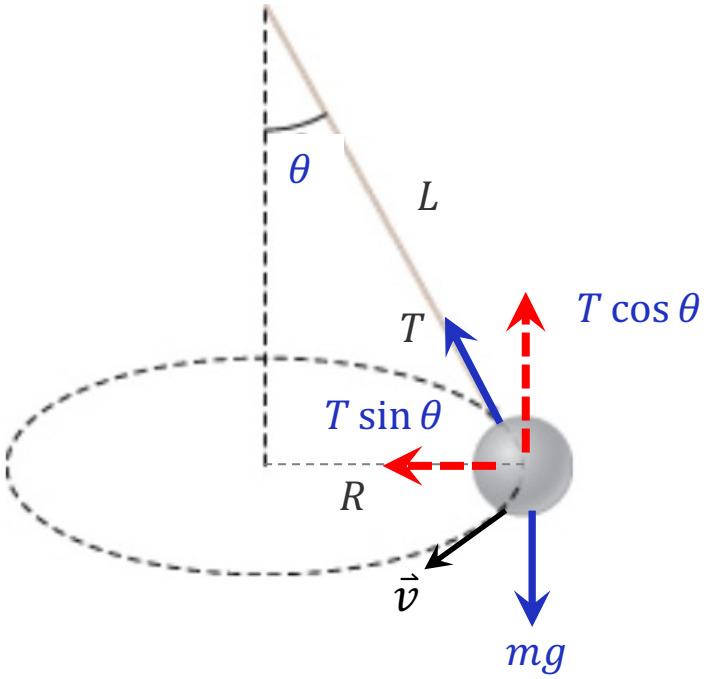
$m = 0.5\ kg$

$R = 1.5\ m$

$T_{max} = 50\ N$

Examples

2) The conical pendulum problem. Find expression for the bob velocity.



No motion in vertical direction.

$$T\cos\theta = m\,g \qquad (1)$$

Centripetal force

$$T\sin\theta = \frac{mv^2}{R} \qquad (2)$$

Divide 2/1

$$\tan\theta = \frac{v^2}{Rg}$$

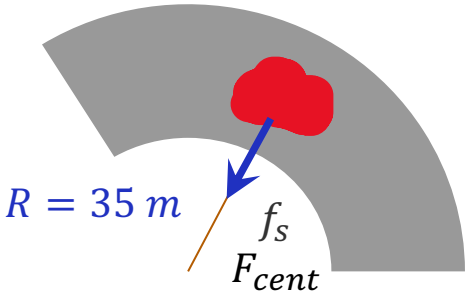
$$\Rightarrow \quad v = \sqrt{R\,g\tan\theta} \quad = \sqrt{L\,g\sin\theta\tan\theta}$$

Examples

3) Car skidding on highway exit. Find maximum speed before skidding.

Car mass $m = 500\ kg$

Static friction $\mu_s = 0.5$



Start of skidding $F_{cent} = \mu_s F_N$

$$\frac{mv^2}{R} = \mu_s mg$$

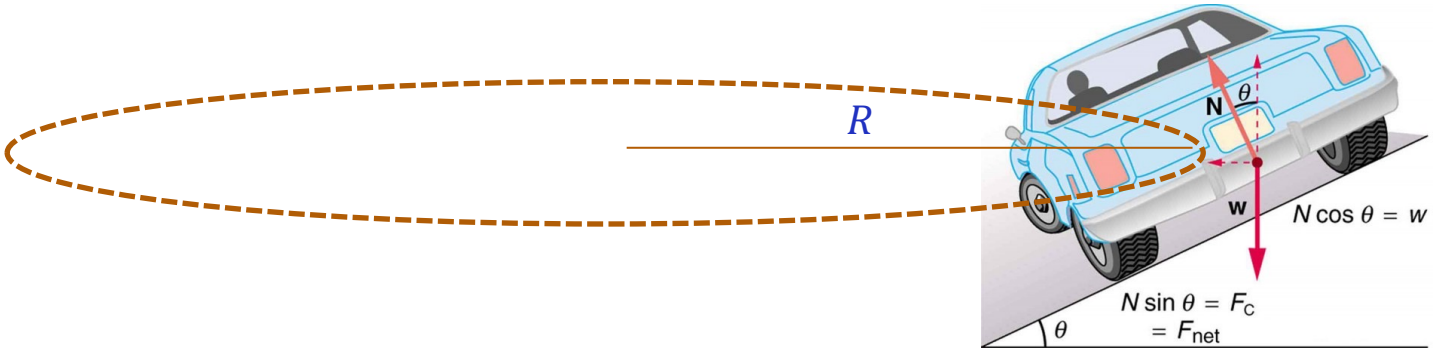
$$v_{max} = \sqrt{\mu_s R g}$$

$$v_{max} = 13.1\ m/s$$

Mass independent!

Examples

4) The highway exit slope design: Banked ramp.



Works even with very low friction.
(worst-case scenario)

Similar to conical pendulum

$$\tan\theta = \frac{v^2}{Rg}$$

For a speed limit of 25 m/s

$$R = 350\ m$$

$$\theta = 10°$$

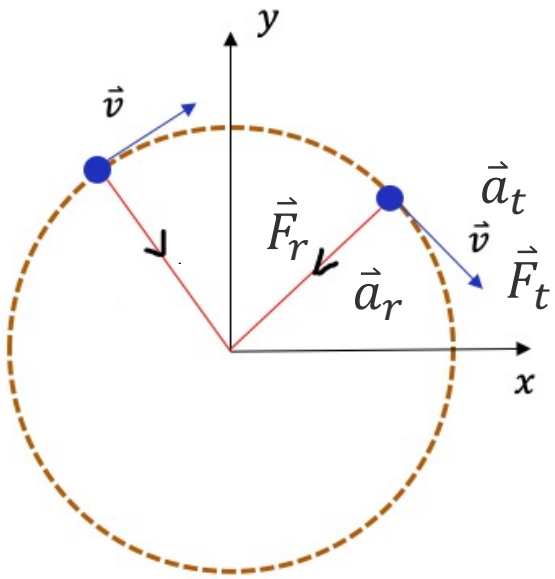
Non-uniform Circular Motion

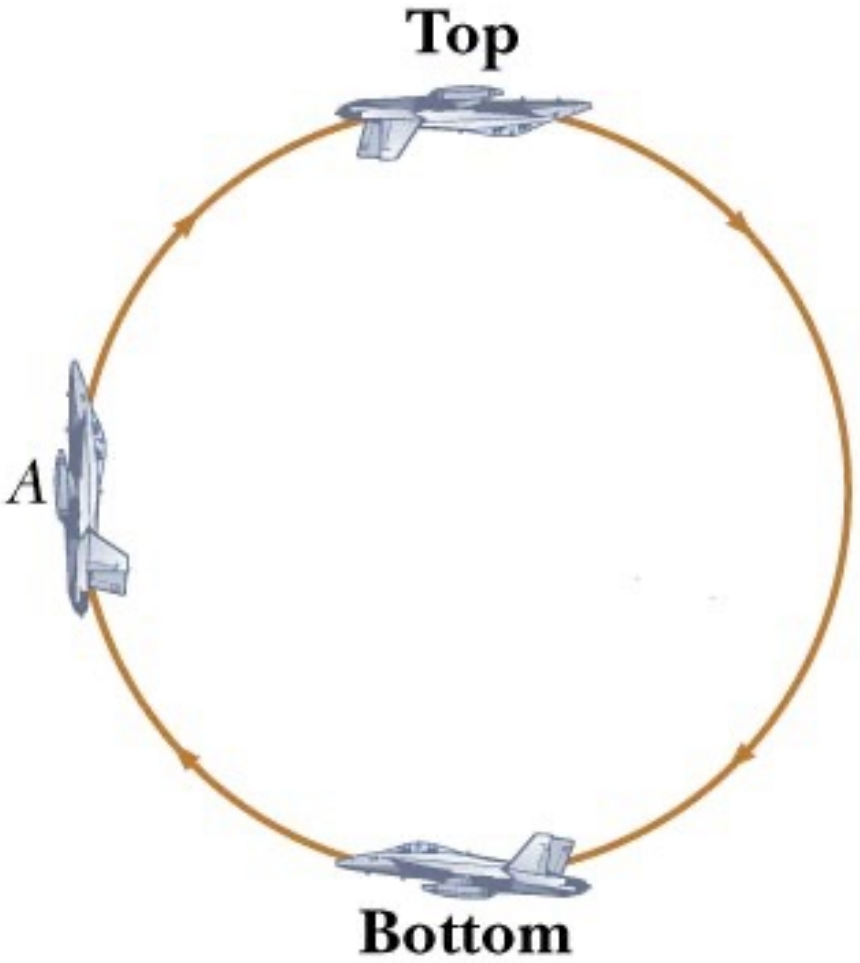
Velocity is not constant and net force is changing.

If $\frac{dv}{dt} \neq 0$, we have

$$\vec{a} = \vec{a}_t + \vec{a}_r$$

$$\vec{F}_{tot} = \vec{F}_t + \vec{F}_r$$





Non-uniform Circular Motion

Example: Spinning ball.

$$F_t = mg\sin\theta = m\,a_t$$

$$F_r = T - mg\cos\theta = m\,a_r = \frac{mv^2}{R}$$

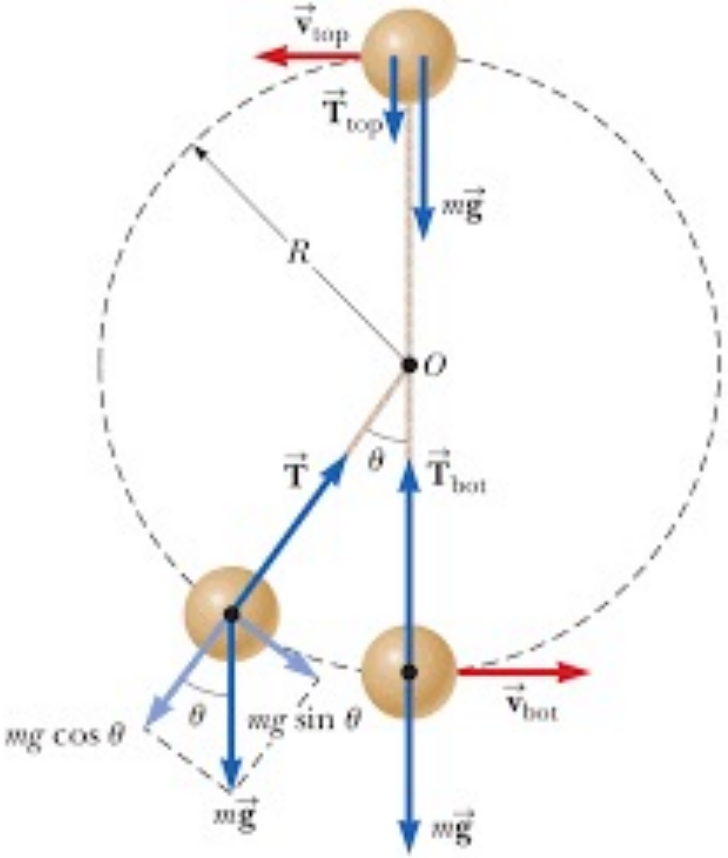
Solve for tension

$$T = m\left(\frac{v^2}{R} + g\cos\theta\right)$$

At the top $\theta = 180°$ $\quad T = m\left(\frac{v^2}{R} - g\right)$

Minimum velocity required at top

$$v = \sqrt{R\,g}$$



At the bot $\theta = 0°$ $\quad T = m\left(\frac{v^2}{R} + g\right)$ Snap!

Q & A

Send me an email if
you have any
questions.